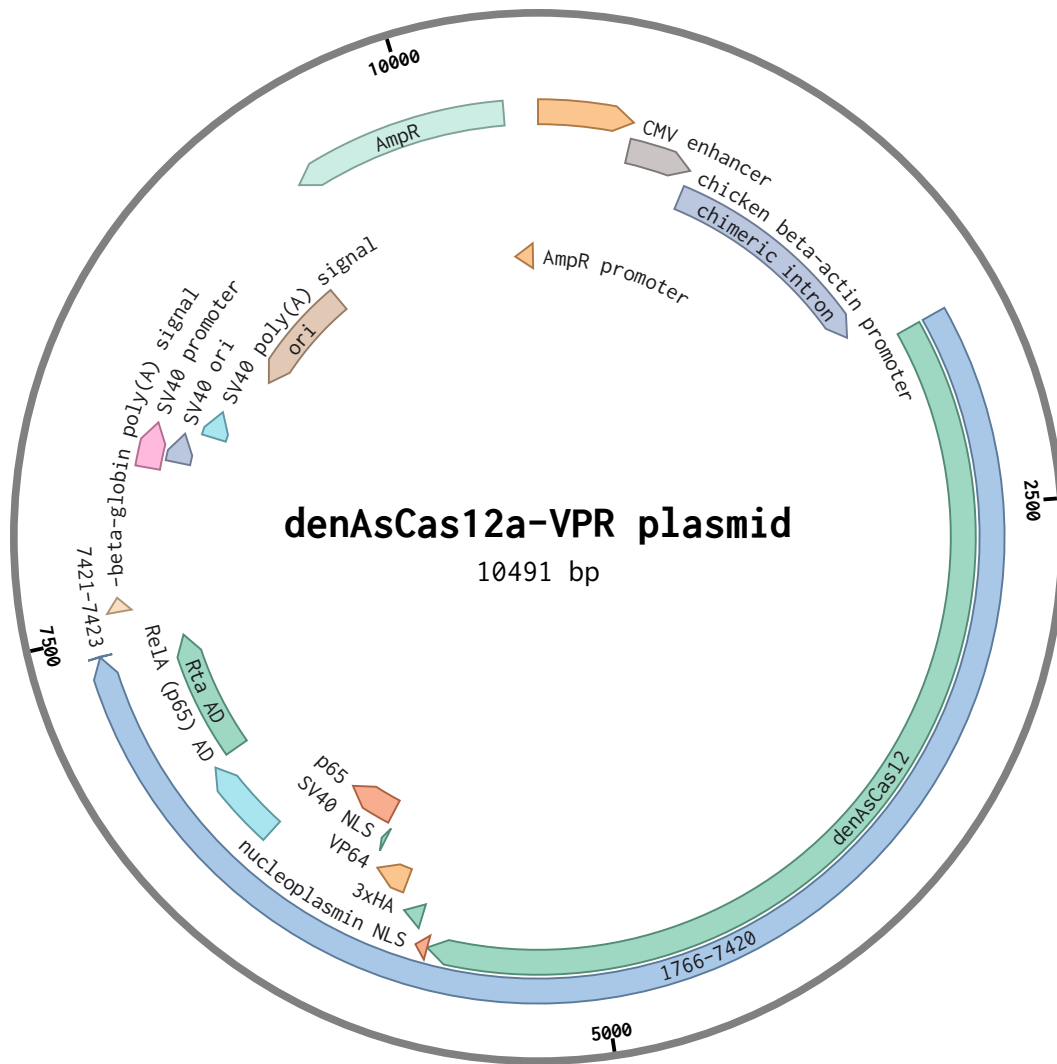


denAsCas12a-VPR plasmid (10491 bp)



(from 1-856 bp)

denAsCas12a-VPR plasmid (10491 bp)

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ctgtaactaataactgatcaataattatcattagttaatgccccagtaatacaagtatcggtatataacctcaaggcgcaatgtattgaatgccatttacggggcgga

CMV enhancer

10 20 30 40 50 60 70 80 90 100

ggctgaccgcccacgacccccgccattgacgtcaataatgacgtatgttcccatagtaacccaataggactttccattgacgtcaatgggtggactatttacg
ccgactggcggggttgctggggcgggtaactgcagttattactgcatacaagggtatcattgcggttatccctgaaaggtaactgcagttaccacactgataaatgc

CMV enhancer

110 120 130 140 150 160 170 180 190 200 210

gtaaactgcccaattggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaattggcccgctggcattatgccagttacatga
catttgacgggtgaaccgtcatgtagttcacatagtagtgcaggttcatgcgggggataactgcagttactgccatttacggggcggaaccgtaatacgggtcatgtact

CMV enhancer

220 230 240 250 260 270 280 290 300 310 320

ccttatgggactttcctacttggcagtacatctacgtattagtcacgcgtattaccatgggtcgaggtgagccccacgttctgcttcaactctccccatctcccccc
ggaataacctgaaaggatgaaccgtcatgtagatgcataatcagtagcgataatggtaccagctccactcggggtgcaagacgaagtgaagggggtagaggggggg

CMV enhancer

chicken beta-actin promoter

330 340 350 360 370 380 390 400 410 420

ctccccaccccccaattttgtattttatttttttaattttttgtgcagcgatggggcgggggggggggggcgcgccagggcgggggcgggggcgagggg
gaggggtgggggttaaaacataataaataaaaaaataataaaacacgtcgctacccccgcccccccccccccgcgcggtccgccccgccccgccccgctcccc

chicken beta-actin promoter

430 440 450 460 470 480 490 500 510 520 530

cggggcgggcgaggcgagaggtgcggcggcagccaatcagagcgcgcgctccgaaagtttccttttatggcgaggcgggcgggcgggcgccctataaaaagcg
gccccgccccgctccgcctctccacgcccgctcggttagtctcgccgcgcgaggctttcaaaggaaaataccgctccgcccgcccgcccggggatatttttcgc

chicken beta-actin promoter

540 550 560 570 580 590 600 610 620 630 640

aagcgcgggcgggcgggagtcgctgcgttgccctcgccccgtgccccgctccgcgcgcctcgcgccccgccccggctctgactgaccgcgttactcccacagg
ttcgcgcgccgccccccctcagcgacgcaacggaagcggggcacggggcgaggcgcgcgagcgcgggcgggcgggcgagactgactggcgcaatgaggggtgtcc

chick...oter

chimeric intron

650 660 670 680 690 700 710 720 730 740

tgagcgggcgggacggcccttctcctccgggtgtaattagcgcttggtttaatgacggctcgtttcttttctgttggtgcgtgaaagccttaaagggtccgggag
actcgccccctgcccgggaagaggaggcccgacattaatcggaaccaaattactgccgagcaaagaaaagacaccgacgcactttcgggaatttcccaggccctc

chimeric intron

750 760 770 780 790 800 810 820 830 840 850

denAsCas12a-VPR plasmid (10491 bp) (from 857-1712 bp)

ggccctttgtcggggggagcggtcgggggtgcgtgcgtgtgtgtgcgtggggagcgccgctgcggcccgctgccggcggtgtgagcgctgcgggcg
ccgggaaacacgccccctcgccgagccccacgcacgcacacacacgcacccctcgcggcgcacgccccggcgacgggcccgcgacactcgcgacgccccg

» chimeric intron »

860 870 880 890 900 910 920 930 940 950 960

cggcgcggggctttgtcgctccgctgtgcgcgaggggagcgcgccggggcggtgccccggtgcgggggggctgcgaggggaacaaaggctgcgtgcgggt
gccgcgccccgaaacacgcgagggcgacacgcgtccccctcgcgccggccccgccacggggcgccacgcccccgacgtccccctgtttccgacgcacgcccc

» chimeric intron »

970 980 990 1,000 1,010 1,020 1,030 1,040 1,050 1,060 1,070

gtgtgcgtgggggggtgagcaggggtgtgggcgcggcggtcgggctgtaacccccctgcacccccctccccgagttgctgagcacggcccggttcgggtgcgg
cacacgcacccccccactcgtccccacacccgcgccgacgcccgcacattggggggggacgtggggggaggggtcaacgactcgtccggggcgaagcccacgcc

» chimeric intron »

1,080 1,090 1,100 1,110 1,120 1,130 1,140 1,150 1,160 1,170

ggctccgtgcggggcggtggcgcggggctcgccgtgccggcggggggtggcggcaggtgggggtgcgggcggggcggggcccgcctcgggcccggggagggctcgggg
ccgaggcacgccccgcaccgcgccccgagcggcacggccccccccaccgcgtccacccccacggcccccgccccggcgagccccgccccctccgagcccc

» chimeric intron »

1,180 1,190 1,200 1,210 1,220 1,230 1,240 1,250 1,260 1,270 1,280

gagggcgcgcgccggccccggagcgccggcggtgtcgaggcgcgcgagccgcagccattgccttttatgtaatcgtgcgagagggcgagggacttcctttgtcc
ctccccgcgcccggggcctcgggcgccgacagctccgcgcgtcggtcggttaacggaaaataccattagcacgctctccgcgtccctgaaggaacaggg

» chimeric intron »

1,290 1,300 1,310 1,320 1,330 1,340 1,350 1,360 1,370 1,380 1,390

caaatctggcggagccgaaatctgggagggcgccgcccacccccctctagcggcgcgggcggaagcggtgcggcgccggcaggaaggaaatgggcggggagggccttc
gtttagaccgcctcggttttagaccctccgcgcgcggtgggggagatcgcccgcgcccgcttcgccacgcccggcggtccttctttaccgcccctccggaag

» chimeric intron »

1,400 1,410 1,420 1,430 1,440 1,450 1,460 1,470 1,480 1,490

gtgcgtgcgcgcccgcgtcccccttccatctccagcctcggggctgcgcagggggacgggtgccttcgggggggacggggcagggcggggttcggcttctggc
cacgcagcgcgcgcgaggggaagaggtagaggtcgagccccgacgggtccccctgccgacggaagccccctgccccgtcccgccccagccgaagaccg

» chimeric intron »

1,500 1,510 1,520 1,530 1,540 1,550 1,560 1,570 1,580 1,590 1,600

gtgtgaccggcggtctagagcctctgtaacctgttcacgttccttcttttctacagctcctgggcaacgtgctggttattgtgctgtctcatcattttggc
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» chimeric intron »

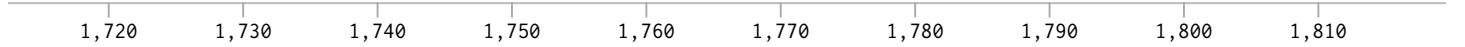
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denAsCas12a-VPR plasmid (10491 bp) (from 1713-2247 bp)

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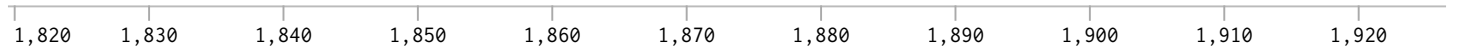
denAsCas12



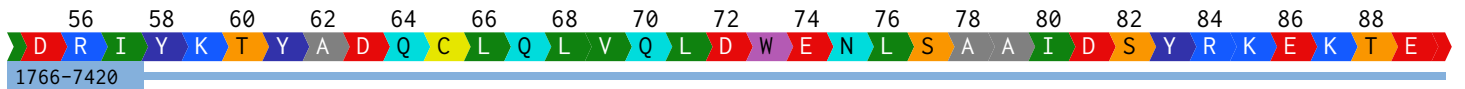
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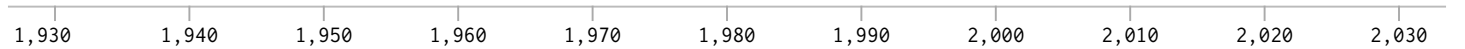
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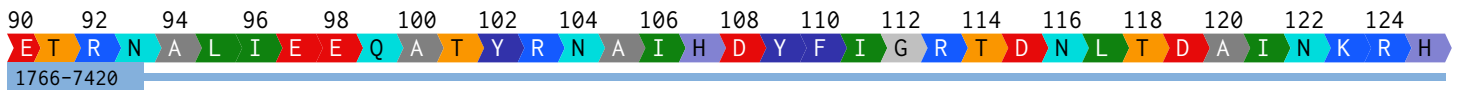
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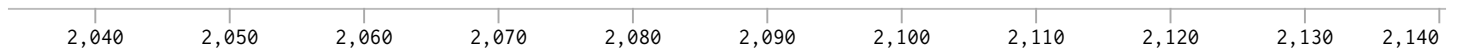
denAsCas12



agacaaggaacgccctgatcgaggagcaggccacatatcgcaatgccatccacgactacttcacgcccggacagacaacctgaccgatgccatcaataagagacac
tctgttccctgcgggactagctcctcgtccggtgtatagcgttacggttaggtgctgatgaagtagccggcctgtctgttgactggctacggttagttattctctgtg



denAsCas12



gccgagatctacaagggcctgttcaaggccgagctgtttaatggcaaggtgctgaagcagctgggcaccgtgaccacaaccgagcacgagaacgcctgctgcggag
cggctctagatgttcccgacaagttccggctcgacaaattaccgttccacgacttcgtcgaccgtggcactgggtgttggtcgtgctcttgcgggacgacgcctc



denAsCas12



denAsCas12a-VPR plasmid (10491 bp) (from 2248-2782 bp)

cttcgacaagttttacaacactacttctccggctttttatagaacaggaagaacgtgttcagcgccgaggatatcagcacagccatccacacgcgcatcgtgcaggaca
gaagctgttcaaatgttgatgaagagccgaaaatatcttgtccttcttgacaaagtcgcggtcctatagtcgtgtcggtagggtgtggcgtagcacgtcctgt

162 164 166 168 170 172 174 176 178 180 182 184 186 188 190 192 194 196
F D K F T T Y F S G F Y R N R K N V F S A E D I S T A I P H R I V Q D

1766-7420
denAsCas12

2,250 2,260 2,270 2,280 2,290 2,300 2,310 2,320 2,330 2,340 2,350

acttcccaagtttaaggagaattgtcacatcttcacacgcctgatcacgcggtgccagcctgcgggagcactttgagaacgtgaagaaggccatcggcacatcttc
tgaaggggttcaaattccttctaacagtgtagaagtggtgcgactagtggtgcgacgggtcggacgccctcgtgaaactcttgacttcttcggtagccgtagaag

198 200 202 204 206 208 210 212 214 216 218 220 222 224 226 228 230 232
N F P K F K E N C H I F T R L I T A V P S L R E H F E N V K K A I G I F

1766-7420
denAsCas12

2,360 2,370 2,380 2,390 2,400 2,410 2,420 2,430 2,440 2,450 2,460

gtgagcacctccatcgaggaggtgttttccttccctttttataaccagctgctgacacagaccagatcgacctgtataaccagctgctgggaggaatctctcgga
cactcgtggaggtagctcctccacaaaaggaagggaataattggtcgacgactgtgtctgggtctagctggacatatgggtcgacgacctccttagagagccct

234 236 238 240 242 244 246 248 250 252 254 256 258 260 262 264 266 268
V S T S I E E V F S F P F Y N Q L L T Q T Q I D L Y N Q L L G G I S R E

1766-7420
denAsCas12

2,470 2,480 2,490 2,500 2,510 2,520 2,530 2,540 2,550 2,560

ggcaggcaccgagaagatcaaggcctgaacgaggtgctgaatctggccatccagaagaatgatgagacagcccacatcatcgctccctgccacacagattcatcc
ccgtccgtggctcttctagttcccgacttgctccacgacttagaccggtaggtcttcttactactctgtcgggtgtagtagcggaggacggtgtgtctaagtagg

270 272 274 276 278 280 282 284 286 288 290 292 294 296 298 300 302
A G T E K I K G L N E V L N L A I Q K N D E T A H I I A S L P H R F I

1766-7420
denAsCas12

2,570 2,580 2,590 2,600 2,610 2,620 2,630 2,640 2,650 2,660 2,670

ccctgtttaagcagatcctgtccgataggaacaccctgtctttcatcctggaggagttaagagcgacgaggaagtgatccagtccttctgcaagtacaagacactg
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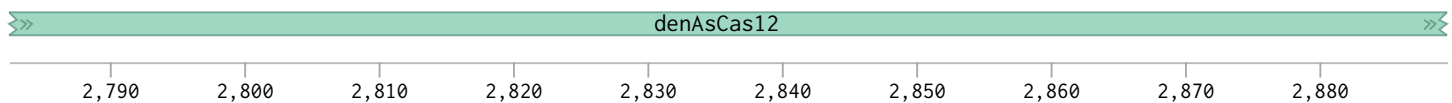
304 306 308 310 312 314 316 318 320 322 324 326 328 330 332 334 336 338
P L F K Q I L S D R N T L S F I L E E F K S D E E V I Q S F C K Y K T L

1766-7420
denAsCas12

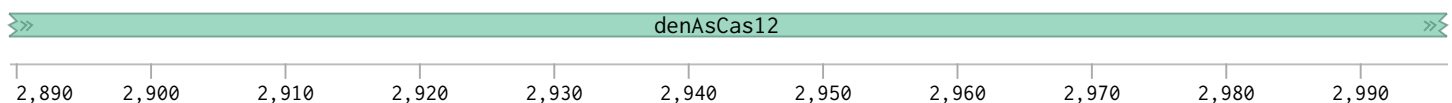
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denAsCas12a-VPR plasmid (10491 bp) (from 2783-3317 bp)

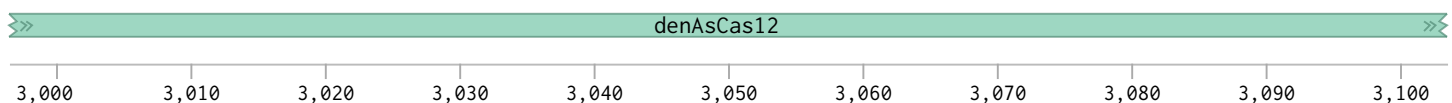
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gactctttgtcttgcacgacctctgtcggtccgggacaaattgctcgacttgcgtagctggactgtgtgtagaagttagtcggtgttcttcgacctctgttagtc
340 342 344 346 348 350 352 354 356 358 360 362 364 366 368 370 372 374
L R N E N V L E T A E A L F N E L N S I D L T H I F I S H K K L E T I S
1766-7420



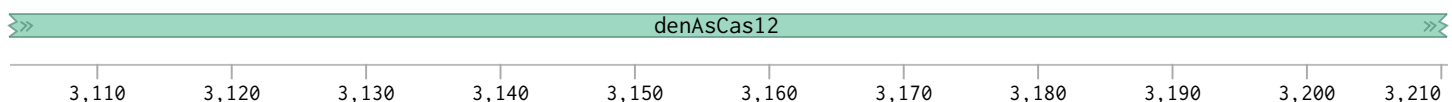
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376 378 380 382 384 386 388 390 392 394 396 398 400 402 404 406 408 410
S A L C D H W D T L R N A L Y E R R I S E L T G K I T K S A K E K V Q
1766-7420



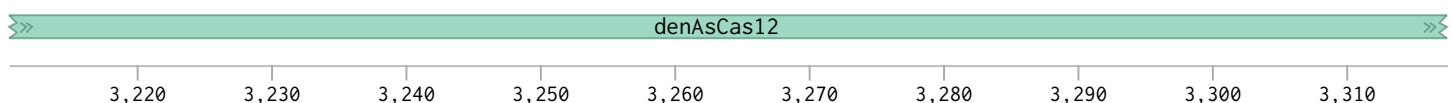
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cgtcgacttcgtgctcctatagttggacgtcctctagtagagacggcgctccgttctctcgactcgctccggaagttcgtcttttggtcgctctaggacagggtgcgt
412 414 416 418 420 422 424 426 428 430 432 434 436 438 440 442 444 446
R S L K H E D I N L Q E I I S A A G K E L S E A F K Q K T S E I L S H A
1766-7420



cacgccgcccctggatcagccactgcctacaacctgaagaagcaggaggagaaggagatcctgaagtctcagctggacagcctgctgggcctgtaccacctgtctgga
gtcggcggggacactagtcggtgacgatgttgggacttcttcgtcctccttctcctctaggacttcagagtcgacctgtcggacgacccggacatggtggacacct
448 450 452 454 456 458 460 462 464 466 468 470 472 474 476 478 480 482
H A A L D Q P L P T T L K K Q E E K E I L K S Q L D S L L G L Y H L L D
1766-7420



ctggtttgccgtggatgagtccaacgaggtggaccccgagtctctgccggctgaccggcatcaagctggagatggagccttctctgagcttctacaacaaggcca
gaccaaacggcacctactcaggttgcctcacctggggctcaagagacgggcccactggccgtagttcgacctctacctcggaagagactcgaagatgttgttccggt
484 486 488 490 492 494 496 498 500 502 504 506 508 510 512 514 516
W F A V D E S N E V D P E F S A R L T G I K L E M E P S L S F Y N K A
1766-7420



denAsCas12a-VPR plasmid (10491 bp) (from 3318-3852 bp)

gaaattatgccaccaagaagccctactccgtggagaagttcaagctgaactttcagatgcctacactggccagaggctgggacgtgaatagagagaagaacaatggc
ctttaatacgggtggttcttcgggatgaggcacctcttcaagttcgacttgaaagtctacggatgtgaccggtctccgacctgcacttatctcttcttgttaccg
518 520 522 524 526 528 530 532 534 536 538 540 542 544 546 548 550 552
R N Y A T K K P Y S V E K F K L N F Q M P T L A R G W D V N R E K N N G
1766-7420

denAsCas12

3,320 3,330 3,340 3,350 3,360 3,370 3,380 3,390 3,400 3,410 3,420

gccatcctgtttgtgaagaacggcctgtactatctgggcatcatgccaaagcagaagggcaggtataagggcctgagcttcgagcccacagagaaaaccagcgaggg
cggtaggacaaacacttcttgcggacatgatagaccgtagtagcttcttccggtccatattccgggactcgaagctcgggtgtctcttttggtcgctccc
554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 588
A I L F V K N G L Y Y L G I M P K Q K G R Y K A L S F E P T E K T S E G
1766-7420

denAsCas12

3,430 3,440 3,450 3,460 3,470 3,480 3,490 3,500 3,510 3,520 3,530

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590 592 594 596 598 600 602 604 606 608 610 612 614 616 618 620 622 624
F D K M Y Y D Y F P D A A K M I P K C S T Q L K A V T A H F Q T H T T
1766-7420

denAsCas12

3,540 3,550 3,560 3,570 3,580 3,590 3,600 3,610 3,620 3,630

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ggtaggacgacaggttgttaaagtagctcggagacctctagtggttctcttagatgctggacttgttaggactcttccctcggttcttcaaagtctgtcggatcggg
626 628 630 632 634 636 638 640 642 644 646 648 650 652 654 656 658 660
P I L L S N N F I E P L E I T K E I Y D L N N P E K E P K K F Q T A Y A
1766-7420

denAsCas12

3,640 3,650 3,660 3,670 3,680 3,690 3,700 3,710 3,720 3,730 3,740

aagaaaaccggcgaccagaagggtacagagaggccctgtgcaagtggatcgacttcacaagggttttctgtccaagtataccaagacaacctctatcgatctgtc
ttcttttggccgctggttcttccgatgtctctccgggacacgttcaccttagctgaagtgtccctaaaagacaggttcatatggttctgttgagatagcttagacag
662 664 666 668 670 672 674 676 678 680 682 684 686 688 690 692 694 696
K K T G D Q K G Y R E A L C K W I D F T R D F L S K Y T K T T S I D L S
1766-7420

denAsCas12

3,750 3,760 3,770 3,780 3,790 3,800 3,810 3,820 3,830 3,840 3,850

denAsCas12a-VPR plasmid (10491 bp) (from 3853-4387 bp)

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atcgagccggttaggagatcatattcctggaccgcgtcatgatacggctcgacttaggggacgacatggtgtagtgaaggctctttagcggctcttccttagt

698 700 702 704 706 708 710 712 714 716 718 720 722 724 726 728 730
S L R P S S Q Y K D L G E Y Y A E L N P L L Y H I S F Q R I A E K E I
1766-7420

denAsCas12

3,860 3,870 3,880 3,890 3,900 3,910 3,920 3,930 3,940 3,950

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acctacggcacctctgtccgttcgacatggacaaggtctagatattgttcctgaaacggttcccgggtggtgccgttcggattagacgtgtgtgacataacctggccg

732 734 736 738 740 742 744 746 748 750 752 754 756 758 760 762 764 766
M D A V E T G K L Y L F Q I Y N K D F A K G H H G K P N L H T L Y W T G
1766-7420

denAsCas12

3,960 3,970 3,980 3,990 4,000 4,010 4,020 4,030 4,040 4,050 4,060

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768 770 772 774 776 778 780 782 784 786 788 790 792 794 796 798 800 802
L F S P E N L A K T S I K L N G Q A E L F Y R P K S R M K R M A H R L G
1766-7420

denAsCas12

4,070 4,080 4,090 4,100 4,110 4,120 4,130 4,140 4,150 4,160 4,170

agagaagatgctgaacaagaagctgaaggatcagaaaaccccaatccccgacaccctgtaccaggagctgtacgactatgtgaatcacagactgtcccacgacctgt
tctcttctacgacttgttcttcgacttcttagtcttttggggtaggggctgtgggacatggtcctcgacatgctgatacacttagtgtctgacaggggtgctggaca

804 806 808 810 812 814 816 818 820 822 824 826 828 830 832 834 836 838
E K M L N K K L K D Q K T P I P D T L Y Q E L Y D Y V N H R L S H D L
1766-7420

denAsCas12

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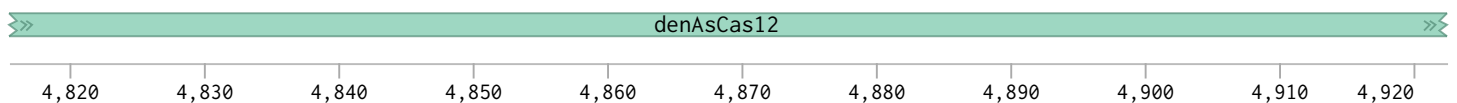
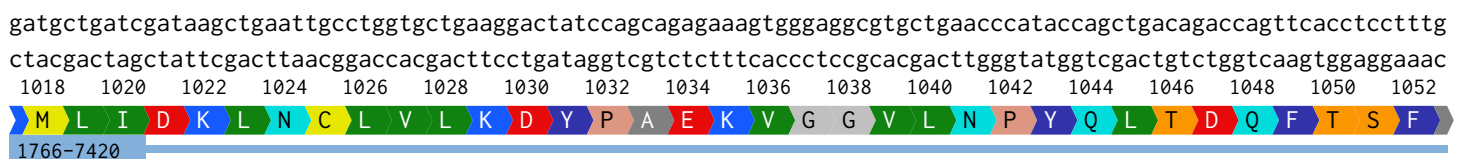
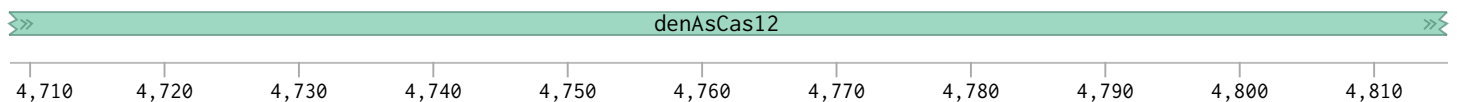
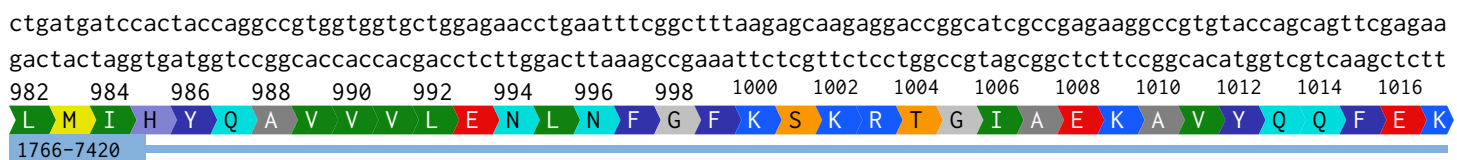
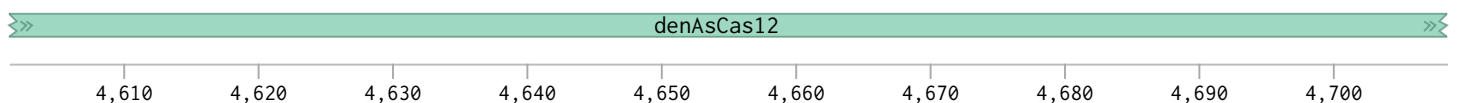
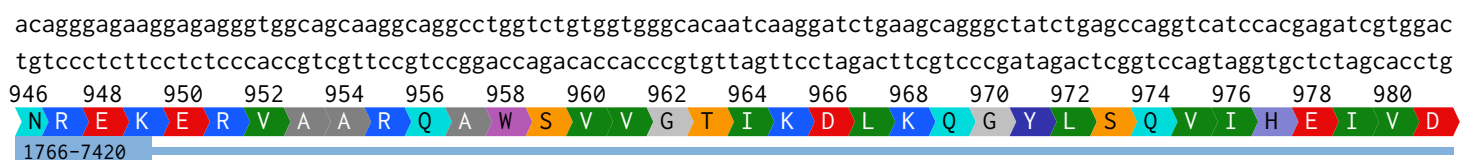
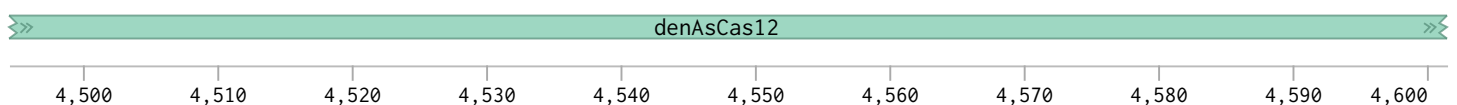
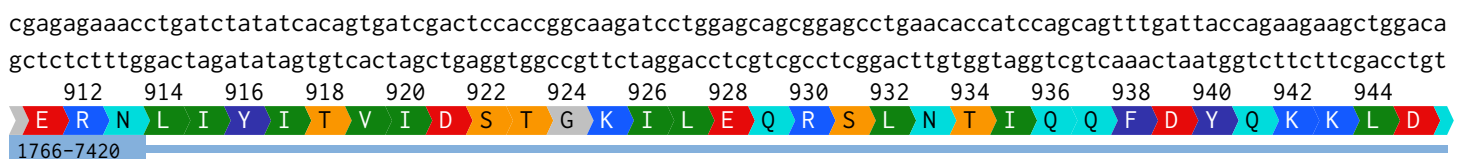
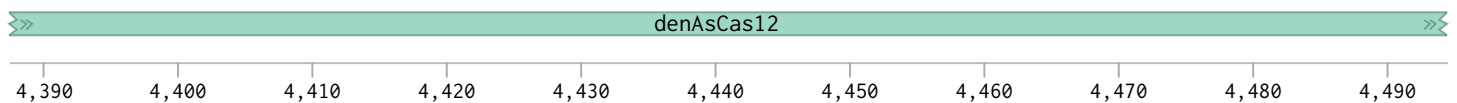
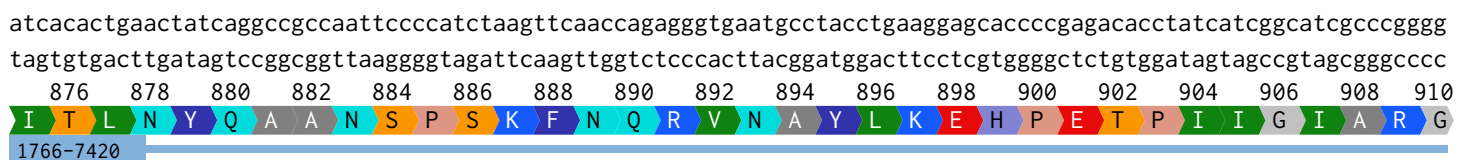
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840 842 844 846 848 850 852 854 856 858 860 862 864 866 868 870 872 874
S D E A R A L L P N V I T K E V S H E I I K D R R F T S D K F F F H V P
1766-7420

denAsCas12

4,290 4,300 4,310 4,320 4,330 4,340 4,350 4,360 4,370 4,380

denAsCas12a-VPR plasmid (10491 bp) (from 4388-4922 bp)



denAsCas12a-VPR plasmid (10491 bp) (from 4923-5457 bp)

ccaagatgggcacccagctctggcttctgtttttacgtgcctgccccatatacatctaagatcgatcccctgaccggcttcgtggacccttcgtgtgaaaaccatc
ggttctaccggtgggtcagaccgaaggacaaaatgcacggacgggtatattgtagattctagctaggggactggccgaagcacctggggaagcacaccttttggtag
1054 1056 1058 1060 1062 1064 1066 1068 1070 1072 1074 1076 1078 1080 1082 1084 1086 1088

A K M G T Q S G F L F Y V P A P Y T S K I D P L T G F V D P F V W K T I

1766-7420

denAsCas12

4,930 4,940 4,950 4,960 4,970 4,980 4,990 5,000 5,010 5,020

aagaatcacgagagccgcaagcacttctggagggttcgactttctgcactacgacgtgaaaaccggcgacttcatcctgcactttaagatgaacagaaatctgtc
ttcttagtgctctcggcgttctgaaggacctccgaagctgaaagacgtgatgctgcacttttggccgctgaagtaggacgtgaaattctacttgtcttttagacag
1090 1092 1094 1096 1098 1100 1102 1104 1106 1108 1110 1112 1114 1116 1118 1120 1122 1124

K N H E S R K H F L E G F D F L H Y D V K T G D F I L G H F K M N R N L S

1766-7420

denAsCas12

5,030 5,040 5,050 5,060 5,070 5,080 5,090 5,100 5,110 5,120 5,130

ctccagagggcctgcccggctttatgcctgcatgggatatcgtgttcgagaagaacgagacacagtttgacgccaagggcaccctttcatcgccggaagagaa
gaaggtctccccggacgggcccgaatacggacgtaccctatagcacaagctcttcttgcctgtgtgtaaactgcggttcccgtggggaagtagcgccgttctctt
1126 1128 1130 1132 1134 1136 1138 1140 1142 1144 1146 1148 1150 1152 1154 1156 1158

F Q R G L P G F M P A W D I V F E K N E T Q F D A K G T P F I A G K R

1766-7420

denAsCas12

5,140 5,150 5,160 5,170 5,180 5,190 5,200 5,210 5,220 5,230 5,240

tcgtgccagtgatcgagaatcacagattcaccggcgatataccgggacctgtatcctgccaacgagctgatcgccctgctggaggagaagggcacgtgttcagggat
agcacggtcactagctcttagtgtctaagtggcgtctatggccctggacataggacggttgctcgactagcgggacgacctcctcttcccgtagcacaagtcctta
1160 1162 1164 1166 1168 1170 1172 1174 1176 1178 1180 1182 1184 1186 1188 1190 1192 1194

I V P V I E N H R F T G R Y R D L Y P A N E L I A L L E E K G I V F R D

1766-7420

denAsCas12

5,250 5,260 5,270 5,280 5,290 5,300 5,310 5,320 5,330 5,340 5,350

ggctccaacatcctgccaagctgctggagaatgacgattctcacgccatcgacacgatggtggccctgatccgcagcgtgctgcagatgcggaactccaatgccgc
ccgaggtttaggacggtttcgacgaccttactgctaagagtgcggtagctgtgctaccaccgggactaggcgtcgacgacgtctacgccttagggttacggcg
1196 1198 1200 1202 1204 1206 1208 1210 1212 1214 1216 1218 1220 1222 1224 1226 1228 1230

G S N I L P K L L E N D D S H A I D T M V A L I R S V L Q M R N S N A A

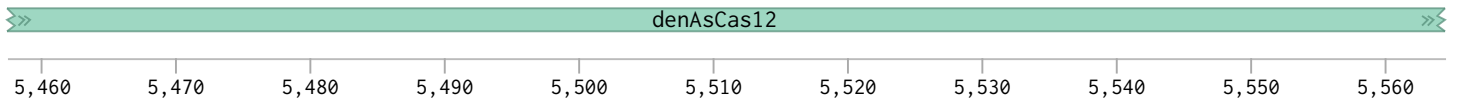
1766-7420

denAsCas12

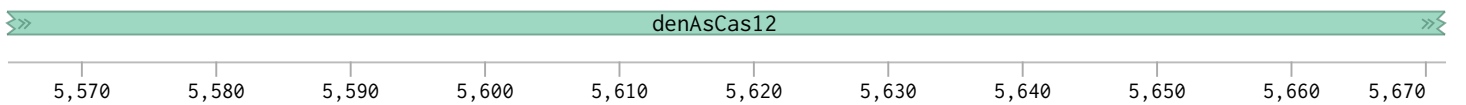
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denAsCas12a-VPR plasmid (10491 bp) (from 5458-5992 bp)

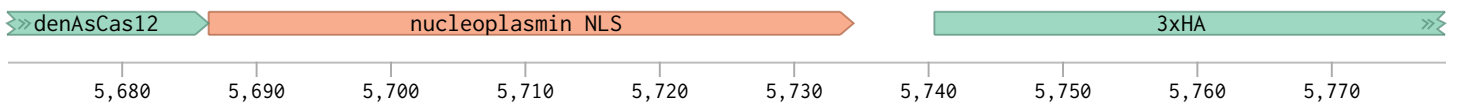
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1232 1234 1236 1238 1240 1242 1244 1246 1248 1250 1252 1254 1256 1258 1260 1262 1264 1266
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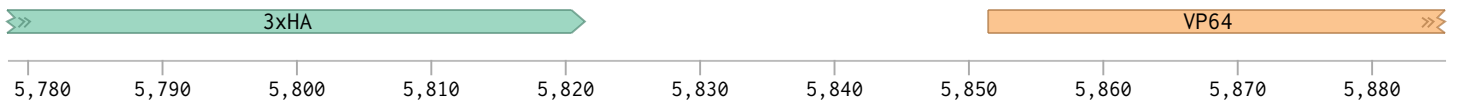
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1268 1270 1272 1274 1276 1278 1280 1282 1284 1286 1288 1290 1292 1294 1296 1298 1300 1302
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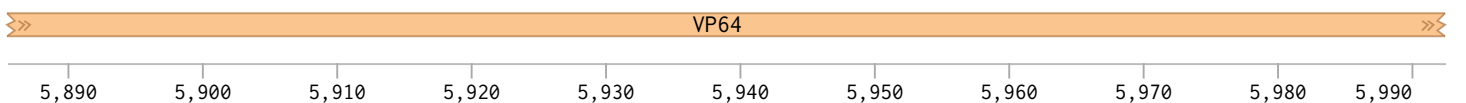
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1304 1306 1308 1310 1312 1314 1316 1318 1320 1322 1324 1326 1328 1330 1332 1334 1336 1338
Q E L R N K R P A A T K K A G Q A K K K K G S Y P Y D V P D Y A Y P Y D



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gcacggactaatacgtatgggtatactacaggggctgatacggccttcgctccggtcgccaaggcctgccgactgcgtaacctgctaaaactagacctatacacc
1340 1342 1344 1346 1348 1350 1352 1354 1356 1358 1360 1362 1364 1366 1368 1370 1372
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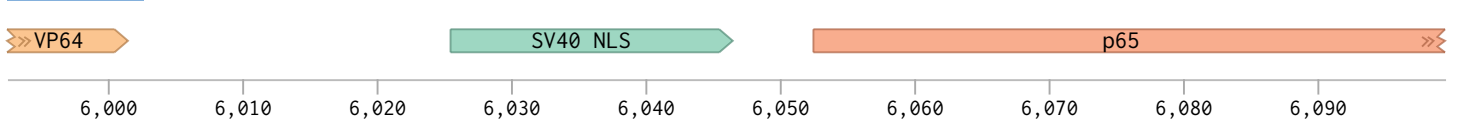
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1374 1376 1378 1380 1382 1384 1386 1388 1390 1392 1394 1396 1398 1400 1402 1404 1406 1408
G S D A L D D F D L D M L G S D A L D D F D L D M L G S D A L D D F D L



denAsCas12a-VPR plasmid (10491 bp) (from 5993-6527 bp)

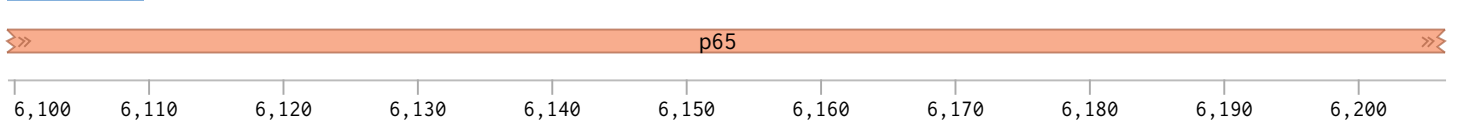
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1410 1412 1414 1416 1418 1420 1422 1424 1426 1428 1430 1432 1434 1436 1438 1440 1442 1444
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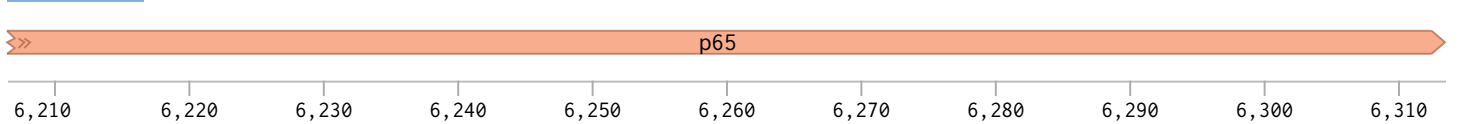
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1446 1448 1450 1452 1454 1456 1458 1460 1462 1464 1466 1468 1470 1472 1474 1476 1478 1480
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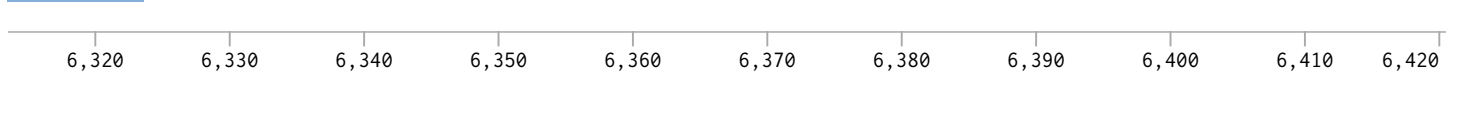
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1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 1508 1510 1512 1514 1516
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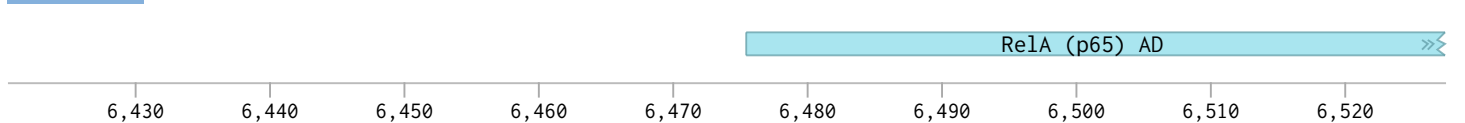
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1518 1520 1522 1524 1526 1528 1530 1532 1534 1536 1538 1540 1542 1544 1546 1548 1550 1552
S Q A S A L A P A P P Q V L P Q A P A P A P A P A M V S A L A Q A P A P



cgtgcctgtgctggctcctggacctccacaggctgtggctccaccagcccctaaacctacacaggccggcgagggcacactgtctgaagctctgctgcagctgcagt
gcacggacacgacccaggacactggaggtgtccgacaccgaggtggtcggggatttggatgtgtccggccgctcccgtgtgacagacttcgagacgacgtcgacgtca

1554 1556 1558 1560 1562 1564 1566 1568 1570 1572 1574 1576 1578 1580 1582 1584 1586
V P V L A P G P P Q A V A P P A P K P T Q A G E G T L S E A L L Q L Q



denAsCas12a-VPR plasmid (10491 bp) (from 6528-7062 bp)

tcgacgacgaggatctgggagccctgctgggaaacagcaccgatcctgccgtgttcaccgacctggccagcgtggacaacagcgagttccagcagctgctgaaccag
agctgctgctcctagaccctcgggacgacctttgtcgtggctaggacggcacaagtggctggaccggctcgacactgttgtcgtcaaggtcgtcagcacttggtc
1588 1590 1592 1594 1596 1598 1600 1602 1604 1606 1608 1610 1612 1614 1616 1618 1620 1622

F D D E D L G A L L G N S T D P A V F T D L A S V D N S E F Q Q L L N Q
1766-7420

RelA (p65) AD
6,530 6,540 6,550 6,560 6,570 6,580 6,590 6,600 6,610 6,620 6,630

ggcatccctgtggccctcacaccaccgagcccatgctgatggaatacccccaggccatcaccggctcgtgacaggcgctcagaggcctcctgatccagctcctgc
ccgtaggacaccggggagtgtggtggctcgggtacgactaccttatggggctcggtagtgggccgagcactgtccgcgagtcctcggaggactaggctcagggacg
1624 1626 1628 1630 1632 1634 1636 1638 1640 1642 1644 1646 1648 1650 1652 1654 1656 1658

G I P V A P H T O T E P M L M E Y P E A I T R L V T G A Q R P P D P A P A
1766-7420

RelA (p65) AD
6,640 6,650 6,660 6,670 6,680 6,690 6,700 6,710 6,720 6,730 6,740

ccctctgggagcaccaggcctgcctaattggactgctgtctggcgacgaggacttcagctctatcgccgatattgatttctcagccttgctgggctctggcagcgga
gggagacctcgtgggtccggacggattacctgacgacagaccgctgctcctgaagtcgagatagcggctatacctaaagagtcggaacgacccgagaccgtcgccgt
1660 1662 1664 1666 1668 1670 1672 1674 1676 1678 1680 1682 1684 1686 1688 1690 1692 1694

P L G A P G L P N G L L S G D E D F S S I A D M D F S A L L G S G S G
1766-7420

RelA (p65) AD
6,750 6,760 6,770 6,780 6,790 6,800 6,810 6,820 6,830 6,840

gccgggattccagggaaggatgttttgcgaagcctgaggccggctccgctattagtacgtgtttgagggccgaggtgtgccagccaaaacgaatccggcca
cggccctaaggctcccttcctacaaaaacggcttcggactcggccgaggcgataatcactgcacaaactccggcgctccacacggctcggttttcttaggcgggt
1696 1698 1700 1702 1704 1706 1708 1710 1712 1714 1716 1718 1720 1722 1724 1726 1728 1730

S R D S R E G M F L P K P E A G S A I S D V F E G R E V C Q P K R I R P
1766-7420

Rta AD
6,850 6,860 6,870 6,880 6,890 6,900 6,910 6,920 6,930 6,940 6,950

tttcatcctccaggaagtccatgggccaaccgcccactccccgccagcctcgaccaacaccaaccggtccagtacatgagccagtcgggtcactgacccccggcacc
aaagtaggaggtccttcaggtaccgggttgccgggtgagggcggtcgagcgtggttgggtggccaggtcatgtactcggtcagcccagtcgactggggccgtgg
1732 1734 1736 1738 1740 1742 1744 1746 1748 1750 1752 1754 1756 1758 1760 1762 1764 1766

F H P P G S P W A N R P L P A S L A P T P T G P V H E P V G S L T P A P
1766-7420

Rta AD
6,960 6,970 6,980 6,990 7,000 7,010 7,020 7,030 7,040 7,050 7,060

denAsCas12a-VPR plasmid (10491 bp) (from 7063-7704 bp)

agtcctcagccactggatccagcgcggcagtcactcccaggccagtcacgtgttgaggatcccgatgaagagacgagccaggctgtcaaagcccttcgggaga
tcaggagatcggtgacctaggtcgcgggcgctcactgagggtccggtcagtggaacacctctagggtacttctctgctcggtccgacagtttcgggaagccctct

1768 1770 1772 1774 1776 1778 1780 1782 1784 1786 1788 1790 1792 1794 1796 1798 1800
V P Q P L D P A P A V T P E A S H L L E D P D E E T S Q A V K A L R E

1766-7420
Rta AD

7,070 7,080 7,090 7,100 7,110 7,120 7,130 7,140 7,150 7,160

tggccgatactgtgattccccagaaggaagaggctgcaatctgtggccaaatggacctttcccatccgcccccaaggggcatctggatgagctgacaaccacactt
accggctatgacactaaggggtcttcttctccgacgttagacaccggtttacctggaaagggtaggcggggttccccgtagacctactcgactgttggtgtgaa

1802 1804 1806 1808 1810 1812 1814 1816 1818 1820 1822 1824 1826 1828 1830 1832 1834 1836
M A D T V I P Q K E E A A I C G Q M D L S H P P P R G H L D E L T T T L

1766-7420
Rta AD

7,170 7,180 7,190 7,200 7,210 7,220 7,230 7,240 7,250 7,260 7,270

gagtcgatgaccgaggatctgaacctggactcaccctgaccccggaattgaacgagattctggataccttctgaacgacgagtgcctcttgcattgcatat
ctcaggtactggctcctagacttgacactgagtggggactggggccttaacttgctctaagacctatggaaggacttgctgctcacggagaacgtacggtacgtata

1838 1840 1842 1844 1846 1848 1850 1852 1854 1856 1858 1860 1862 1864 1866 1868 1870 1872
E S M T E D L N L D S P L T P E L N E I L D T F L N D E C L L H A M H I

1766-7420
Rta AD

7,280 7,290 7,300 7,310 7,320 7,330 7,340 7,350 7,360 7,370 7,380

cagcacaggactgtccatcttcgacacatctctgttttaagcgccgactcctcagggtgcaggctgcctatcagaagggtggtggtgtggccaatgccctggc
gtcgtgtcctgacaggtagaagctgtgttagacacaaaattcgccggcgtgaggagtccacgtccgacgtagtcttccaccaccgaccacaccggttacgggaccg

1874 1876 1878 1880 1882 1884 1
S T G L S I F D T S L F *

1766-7420
Rta AD

7,390 7,400 7,410 7,420 7,430 7,440 7,450 7,460 7,470 7,480 7,490

tcacaaataccactgagatcttttccctctgcaaaaattatggggacatcatgaagcccttgagcatctgacttctggctaataaaggaaatttattttcattg
agtgtttatggtgactctagaaaaaggagacggtttttaataccctgttagtacttcggggaactcgtagactgaagaccgattatttcttttaataaaagtaac

-beta-glo...) signal

7,500 7,510 7,520 7,530 7,540 7,550 7,560 7,570 7,580 7,590

caatagtgtgttggaatTTTTGTGTCTCTCACTCGGAAGGACATATGGGAGGGCAAATCATTTAAACATCAGAAATGAGTATTTGGTTTAGAGTTTGGAACATAT
gttatcacacaaccttaaaaaacagagagttagccttctgtataccctcccgtttagtaaattttgtagtcttactcataaaccaaatctcaaaccgttgtata

-beta-globin poly(A) signal

7,600 7,610 7,620 7,630 7,640 7,650 7,660 7,670 7,680 7,690 7,700

denAsCas12a-VPR plasmid (10491 bp) (from 7705-8667 bp)



denAsCas12a-VPR plasmid (10491 bp) (from 8668-9523 bp)

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tcctttctgtacactcggtttccgggtcggtttccgggtccttggcattttccggcgcaacgaccgcaaaaagggtatccgagggcggggggactgctcgtagtgtttt

ori

8,670 8,680 8,690 8,700 8,710 8,720 8,730 8,740 8,750 8,760 8,770

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tagctgcgagttcagctctccaccgctttgggctgtcctgatatttctatggtccgcaaagggggaccttcgagggagacgcgagaggacaaggctgggacggcgaa

ori

8,780 8,790 8,800 8,810 8,820 8,830 8,840 8,850 8,860 8,870 8,880

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ori

8,890 8,900 8,910 8,920 8,930 8,940 8,950 8,960 8,970 8,980

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acacgtgcttggggggcaagtcgggctggcgacgcggaataggccattgatagcagaactcaggttgggccattctgtgctgaatagcggtgaccgtcgtcggtgac

ori

8,990 9,000 9,010 9,020 9,030 9,040 9,050 9,060 9,070 9,080 9,090

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ori

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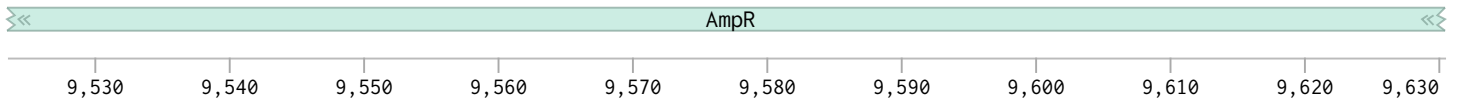
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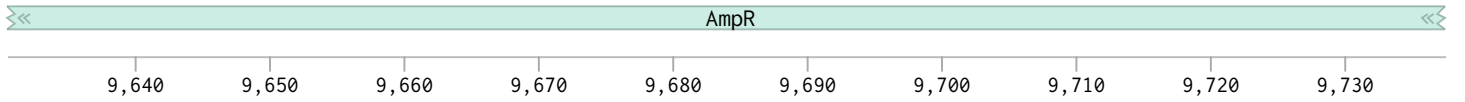
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denAsCas12a-VPR plasmid (10491 bp) (from 9524-10379 bp)

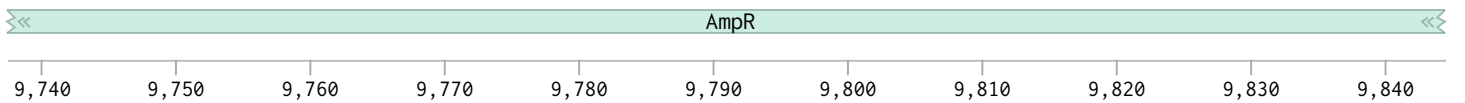
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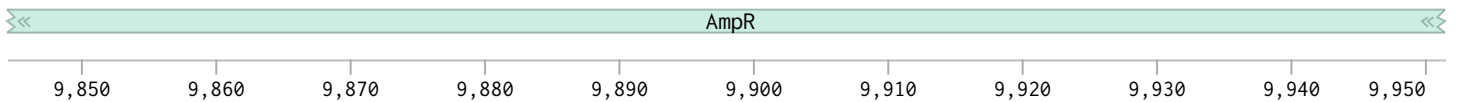
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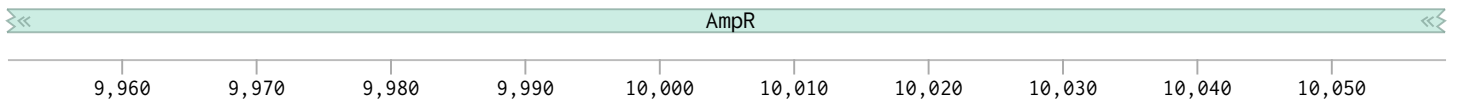
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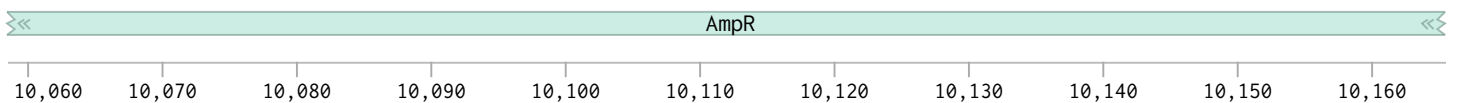
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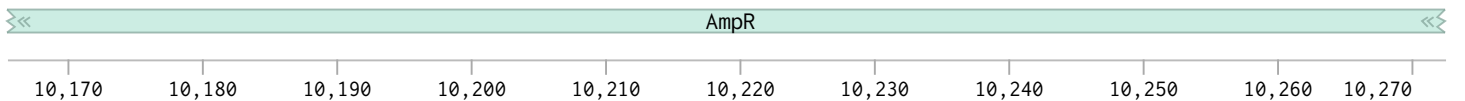
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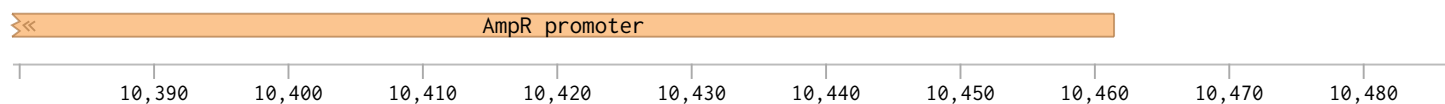


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denAsCas12a-VPR plasmid (10491 bp) (from 10380-10491 bp)

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